AI-103
Azure AI App and Agent Developer Trainer Preparation Guide

March 2026

Purpose

This document is for Microsoft Certified Trainers preparing to teach AI-103: Developing Azure AI Apps and Agents. This course is designed for students who are planning to take the corresponding certification exam, or students who are performing Azure AI App and Agent Developer tasks in their daily job.

Course description

AI-103: Developing Azure AI Apps and Agents is designed for software developers building production-ready AI solutions on Microsoft Azure. This course covers the full spectrum of development with Microsoft Foundry—from deploying and optimizing generative AI models and implementing RAG pipelines, to building autonomous agents with tool integration, multi-agent orchestration, and responsible AI safeguards. Students also explore multimodal capabilities including vision, speech, natural language analysis, and document/content extraction, learning to connect knowledge sources and external services into intelligent agentic workflows.

Learning objectives

After completing this course, students will be able to:

- Build generative AI applications using Microsoft Foundry, including RAG pipelines, prompt optimization, and responsible AI practices
- Develop AI agents that integrate tools, memory, MCP, and multi-agent orchestration using Azure AI Agent Service and the Microsoft Agent Framework
- Implement multimodal solutions that process and generate images, video, speech, and complex documents
- Apply text analysis and natural language solutions using Azure AI Language services and generative AI capabilities in Foundry
- Build information extraction and knowledge grounding pipelines using Azure AI Search, Content Understanding, and document intelligence

Azure AI App and Agent Developer role definition

Both the certification exam and the courseware are based on the Azure AI App and Agent Developer role.

Azure AI App and Agent Developers build, deploy, and manage AI solutions using Microsoft Azure. They are responsible for developing generative AI applications, autonomous agents, and multimodal workflows that leverage Microsoft Foundry and Azure AI services. Azure AI App and Agent Developers collaborate with data scientists and software developers to integrate AI capabilities into production systems, ensuring solutions are secure, scalable, and aligned with responsible AI principles.

Audience profile

This course is intended for software developers wanting to build AI infused applications that leverage Microsoft Foundry. Topics in this course include developing generative AI apps, building AI agents, and solutions that implement knowledge connections or tools in your agentic applications. This course also covers multimodal capabilities and understanding of complex content. This course was designed for software engineers concerned with building, managing and deploying AI solutions that leverage Microsoft Foundry. They are familiar with Python and have knowledge on using APIs and SDKs to build agents and generative AI solutions on Azure.

Audience prerequisites

Before attending this course, students should have: familiarity with Python programming; experience using REST APIs and SDKs to build cloud-connected applications; foundational knowledge of Azure services and the Azure portal; and an understanding of basic AI/ML concepts such as machine learning models and inferencing. Prior experience with Azure AI services is helpful but not required.

Certification exam

The Microsoft Certified: Azure AI App and Agent Developer Associate Certification aligns with today's generative and agentic architectures for planning and managing AI resources in Microsoft Foundry, building generative apps and multistep reasoning workflows, and developing production-ready agents with multi-agent orchestration capabilities. Exam AI-103 beta available in April 2026; exam expected to go live in June 2026.

The study areas for the certification exam related to this course are based on the Job Task Analysis (JTA) that was conducted in March 2026.

Each study area has a percentage indicating the relative weight of the area on the exam. The higher the percentage, the more questions a candidate is likely to see in that area.

To review the full objective domain the skills measured, review the AI-103 study guide.

Study Area	Percentage
Plan and manage an Azure AI solution	25-30%
Implement generative AI and agentic solutions	30-35%
Implement computer vision solutions	15-20%
Implement text analysis solutions	10-15%
Implement information extraction solutions	10-15%

For exams related to non-Fundamental courses, candidates should have a minimum of six months of hands-on experience working with Azure AI and Foundry (AI-103).

Prerequisite knowledge to teach this course

To successfully teach this course, instructors must have a solid working knowledge of Microsoft Azure AI services and the Microsoft Foundry SDK. They should have practical experience developing and deploying generative AI solutions, including RAG pipelines, AI agents, and multi-agent orchestration patterns.

Note: These guidelines are not inclusive of the requirements to become a Microsoft Certified Trainer (MCT).

Required materials to prepare for and teach this course

You need the following materials to prepare for and teach this course:

Resource	Description
Microsoft PowerPoint files	Download the AI-103T00-ENU-PowerPoint.zip from the MCT Download Center.
Change Log	Download the AI-103T00-ENU-ChangeLog.pdf from the MCT Download Center.
Lab environment provided by your lab hosting provider	Contact your lab hosting provider for instructions on using their lab environment.
Lab instructions	The lab instructions are provided in the lab environment and in the following Microsoft Learning GitHub repositories: LP1 (Foundry): mslearn-ai-studio LP2 (Agents): mslearn-ai-agents LP3 (Language and Speech): mslearn-ai-language

	LP4 (Vision and Info Extraction): mslearn-ai-vision, mslearn-ai-information-extraction
Student training content	See the following section for a detailed breakdown of each Learning Path covered in the course.
Microsoft Learn for Educators program	For educational institutions that deliver this course, additional assets are available through the Learning Download Center (LDC) for Educators.

Student training content

The student training content for this course is located in Microsoft Learn. The following table provides a breakdown of each Learning Path, the modules covered in each, and the link to each LP in Microsoft Learn.

The student training content includes links to additional reading material to help you prepare for specific topic areas.

Learning Path	Online training in Microsoft Learn
Course Introduction	Slides only
Learning Path 1: **Develop generative AI apps with Microsoft Foundry**	**Module 1:** Plan and prepare to develop AI solutions on Azure **Module 2:** Evaluate models with the Azure AI Foundry model catalog **Module 3:** Develop an AI app with the Microsoft Foundry SDK **Module 4:** Optimize generative AI model performance **Module 5:** Implement a responsible generative AI solution in Microsoft Foundry
Learning Path 2: **Develop AI agents with Azure**	**Module 1:** Develop AI agents with Microsoft Foundry and Visual Studio Code **Module 2:** Build an agent with custom tools **Module 3:** Connect an agent to MCP tools **Module 4:** Build knowledge-enhanced AI agents with Foundry IQ **Module 5:** Integrate your agent with M365 **Module 6:** Build agent-driven workflows using Microsoft Foundry **Module 7:** Develop an AI agent with the Microsoft Agent Framework **Module 8:** Orchestrate a multi-agent solution using the Microsoft Agent Framework **Module 9:** Discover Azure AI Agents with A2A

Learning Path 3: **Develop language solutions with Azure AI**	**Module 1:** Analyze text with Azure AI Language **Module 2:** Develop a text analysis agent with Language MCP **Module 3:** Develop generative AI audio apps **Module 4:** Create speech-enabled apps with Microsoft Foundry **Module 5:** Develop a speech agent with Azure AI Speech MCP **Module 6:** Develop a voice live agent **Module 7:** Translate text and speech
Learning Path 4: **Extract insights from visual data on Azure**	**Module 1:** Generate images with AI **Module 2:** Generate and edit video with AI **Module 3:** Analyze media with Azure Content Understanding **Module 4:** Develop a vision-enabled generative AI application **Module 5:** Create a multimodal analysis solution with Azure Content Understanding **Module 6:** Create an Azure Content Understanding client application **Module 7:** Create a knowledge mining solution with Azure AI Search

Preparation tasks

Instructors should complete the following tasks to prepare for teaching this course:

- If you have previously taught this class, refer to the course's Change Log. It provides detailed information on how the course has changed over time. The Change Log is updated for each course release.
- Review all topics in the student training material in Microsoft Learn (see the link in the Required Materials section above). You should be well-versed in every topic. If you have previously taught the course and are comfortable with your knowledge of each topic, focus primarily on the new or updated topics as outlined in the Change Log.
- Review the PowerPoint slides.
    - Be able to speak to each of the talking points on the slides. Some slides include a graphic from the associated Learn content for the topic. These graphics are provided on the slide so that you can speak to them to help explain the key talking points in the topic.
    - **The bulleted items on each slide should NOT be read verbatim to the students.** The students can read the slides themselves. Rather, the bullet points reflect the key information that you should focus on when discussing each topic. You should use your experience as a subject matter expert to explain the What, the Why, and the How of each topic. **This is your opportunity to provide a real value-add above and beyond the bulleted talking points.**

- Review the Additional Reading links and other linked resources provided in the student training material. **It's recommended that you present key points from this material to supplement the value-add you provide as an instructor.**
- As you prepare for the class, you should review each unit and determine which ones you want to perform demonstrations of the corresponding product functionality. It's up to you to decide which product features you want to demonstrate to the class. You should use your experience to identify key points during the demonstration process. **This is an area where you should rely on your experience as a subject matter expert to provide additional value-add to the students.**
- You should review each Knowledge Check (KC) question so that you know why the correct answer is correct for each question. Students may challenge some of the questions, so you must be able to address any of those concerns.
- You should perform the labs yourself prior to class so that you become familiar with them and with any of the difficult points in the lab exercises. This will prepare you for helping students in case they get stuck.

Course timing

Daily agenda

Use the table format as presented below. Please make note of the following requirements:

- The agenda items should be in the order in which they will be presented each day.
- Define the total estimated time to complete each agenda item (for example, 30 minutes, 1 hour, etc.).
- Do NOT include start and end times (i.e. 8:00 am to 8:45 am). Not every instructor will start at the same time, nor will they have the same breaks, lunch times, etc. Plus, hybrid and virtual courses may not be taught in the same 8 am to 5 pm cadence as a traditional classroom course.
- Do NOT include lunch and break times as they may vary from one course to the next, especially with hybrid and virtual courses. Just include the modules, labs, and any other classroom activities.>

The following agenda provides estimated times to complete each classroom activity. However, the estimated times may vary depending on the background of your students, which may affect whether you can move faster or slower through the course material.

Estimated times for each Module include the time to complete as part of a 4 day course:

- The module's PowerPoint slide deck presentation.
- Any pre-defined product demonstrations.
- Time to review Knowledge Check questions (see the section on Additional Timing Notes below).
- Time to complete a classroom discussion activity if a Discussion slide is included in the module slide deck.

For courses taught in a longer format such as part pf the Microsoft Learn for Educators (MSLE) program, review the **Design your syllabus** section of the **Educator Teaching Guide** found on the Learning Download Center.

You should adjust the agenda accordingly based on any classroom activities that you personally created or plan to deliver that are not included in the slides for this course. For example, if you plan to present:

- ad-hoc demonstrations
- review activities
- classroom games
- and so on…

Note: Each Learning Path/Module activity in the following agenda is the slide deck presentation for that module.

The following table provides delivery specifications for a 4-day course. For courses taught in a longer format such as part pf the MSLE program, review the **Design your syllabus** section of the **Educator Teaching Guide** found on the Learning Download Center.

Day	Estimated Time	Classroom activity
1	30 minutes	Module 1: Plan and prepare to develop AI solutions on Azure
	60 minutes	Module 2: Evaluate models with the Azure AI Foundry model catalog
	60 minutes	Lab: Model catalog and evaluation lab
	60 minutes	Module 3: Develop an AI app with the Microsoft Foundry SDK
	60 minutes	Lab: Foundry SDK app lab
	60 minutes	Module 4: Optimize generative AI model performance (RAG and prompt engineering)
	60 minutes	Lab: RAG / optimization lab
	30 minutes	Module 5: Implement a responsible generative AI solution in Microsoft Foundry
2	45 minutes	Module 1: Develop AI agents with Microsoft Foundry and Visual Studio Code
	60 minutes	Lab: Core agent build lab
	45 minutes	Module 2: Build an agent with custom tools
	45 minutes	Lab: Tool integration lab

	30 minutes	Module 3: Connect an agent to MCP tools
	45 minutes	Lab: MCP tools lab
	30 minutes	Module 4: Build knowledge-enhanced AI agents with Foundry IQ (demo)
	30 minutes	Module 5: Integrate your agent with M365 (demo)
	45 minutes	Modules 6–8: Agent workflows, Agent Framework, and multi-agent orchestration (overview)
	45 minutes	Lab: Orchestrated multi-agent lab
	15 minutes	Module 9: Discover Azure AI Agents with A2A (awareness)
3	45 minutes	Module 1: Analyze text with Azure AI Language
	60 minutes	Lab: Text analysis lab
	45 minutes	Module 2: Develop a text analysis agent with Language MCP
	45 minutes	Lab: Text analysis agent lab
	30 minutes	Module 3: Develop generative AI audio apps
	45 minutes	Module 4: Create speech-enabled apps with Microsoft Foundry
	45 minutes	Lab: Speech app lab
	30 minutes	Module 5: Develop a speech agent with Azure AI Speech MCP (lab/demo)
	20 minutes	Module 6: Develop a voice live agent (demo)
	30 minutes	Module 7: Translate text and speech
	30 minutes	Lab: Translation lab
4	30 minutes	Module 1 (LP4): Generate images with AI

	45 minutes	Lab: Image generation lab
	30 minutes	Module 2 (LP4): Generate and edit video with AI (demo)
	45 minutes	Module 3 (LP4): Analyze media with Azure Content Understanding
	45 minutes	Lab: Content Understanding media lab
	30 minutes	Module 4 (LP4): Develop a vision-enabled generative AI application
	45 minutes	Modules 1–2 (LP4): Create multimodal analysis solutions with Azure Content Understanding
	45 minutes	Lab: Content Understanding extraction lab
	30 minutes	Modules 3–4 (LP4): Knowledge mining with Azure AI Search and document intelligence
	45 minutes	Lab: AI Search and ingestion pipeline lab
	30 minutes	Course wrap-up, exam prep guidance, and next steps

Additional timing notes - Knowledge check questions

Knowledge check (KC) questions are provided throughout the course to check the student's knowledge of the material that was covered. Instructors can use these KC questions in several ways:

- Conduct a formal classroom exercise in which you go through the questions in a module before moving on to the next module.
- Sprinkle the questions into the content as you cover the related material for a module
- Let the students review the questions after class as a daily homework assignment. You can set aside time at the start of each day to answer any questions they have regarding the prior day's questions. This may be the most feasible option given the tight time constraints that most classes work under.

It will be left up to each instructor to determine how they want to incorporate the KC questions into their class.

If you provide students with time to review the KC questions at the end of specific topics and at the end of each module, you should provide a couple of minutes per question, along with a few extra minutes per question to respond to student questions or challenges concerning certain questions they may not understand or whose answers they disagree with. This may add an extra 15 to 30 minutes to complete each module.

Additional timing notes – Classroom discussions

This course includes two types of classroom discussion questions:

- Discussion questions that are interspersed throughout modules (i.e. a slide that asks a discussion question on a product feature. For example: "Do you see your organization implementing this feature? Why or why not?")

  You should plan on 5-10 minutes for each of these discussion slides.

- A slide at the end of each learning path slide deck that asks students to discuss what they learned in the LP (the final module in each LP is titled Learning Path Review and only contains the discussion slide; this module is a classroom activity and is not associated with a Learn module).

  Each LP Review slide asks the following questions: "What are your key takeaways from this Learning Path? What are the key features discussed in this Learning Path that you foresee implementing at your organization?"

  You should plan on 10-15 minutes to discuss the review slide in the final module of each Learning Path slide deck.

Labs

NOTE: Labs will be available on 4/3/26

The labs must be completed within the lab environment provided by your lab hosting provider. Detailed, step-by-step instructions are provided for each lab and presented as part of the UI experience within your lab environment.

At the time the courses were released, the lab instruction had been thoroughly tested and the lab steps were 100% accurate. However, given the nature of Microsoft's cloud products and the fact that Microsoft releases UI updates on a regular basis, it's possible that at some point in time, the UI for a given feature may change so that it no longer matches the lab instruction.

If students encounter lab steps that don't accurately reflect the UI, they'll have to work through the UI to determine what needs to be done. Typically, UI changes are quite subtle, so hopefully you don't find yourself in a situation where a feature was completely overhauled.

However, if you do run into major UI changes, challenge your students to work through it, and only offer help if they definitely need it. Product UI changes will be part of their daily life in today's cloud-centric world. As IT/Pros, they must learn how to work through such situations.

One thing Microsoft does ask of you is that if you run into situations such as this where lab instructions no longer match the corresponding UI, please document the issue in the course's GitHub repository. This will help Microsoft's World-Wide Learning team update the lab instructions to keep them as up to date as possible. For information on how to submit an issue, please see GitHub User Guide for MCTs.

Potential lab issues

This section identifies lab exercises in which students have run into problems in the past due to the complexity of the lab and/or the students lack of prerequisite background knowledge.

Lab and Exercise	Task	Description
Lab TBD, Exercise TBD TBD	Task TBD TBD	No known issues at this time.
Lab TBD, Exercise TBD TBD	Task TBD TBD	No known issues at this time.
Lab TBD, Exercise TBD TBD	Task TBD TBD	No known issues at this time.
And so on….		

Portal, Cloud Shell, PowerShell, and the CLI

The lab instructions are written to use the Cloud Shell. The Cloud Shell automatically connects to Azure and provides access to PowerShell and the CLI.

If you would rather have students use PowerShell or the CLI locally, you can use these links.

- Install Azure PowerShell on Windows with PowerShellGet
- Install Azure CLI on Windows

Tips and tricks for teaching AI-103

AI-103 is a new course launching in March 2026. As such, no instructor tips and tricks are available yet. This section will be updated with practical guidance based on trainer feedback collected over the coming months. If you have tips or insights to share following your first delivery, please reach out to the course team.

Common product misconceptions

Azure AI Foundry vs. Microsoft Foundry: naming and portal experience

Azure AI Foundry is the previous name for what is now called Microsoft Foundry. All course content has been developed for the new Microsoft Foundry experience.

External documentation: Much of the existing Microsoft documentation still refers to “Azure AI Foundry” or describes the classic portal experience. When sharing links or reference material with students, verify that the content reflects the new Microsoft Foundry interface rather than the classic version, as navigation, terminology, and available features may differ significantly.

In the portal: When students open the Microsoft Foundry portal, they may be prompted to switch between the classic and the new portal experience. Students should always select and work in the new

portal experience when following course labs and exercises. Instructors should confirm this at the start of each lab session to avoid confusion caused by interface differences.

Evolving from AI-102

AI-103 is the successor to AI-102: Designing and Implementing a Microsoft Azure AI Solution. If you have experience teaching AI-102, much of that preparation will carry over. The core Azure AI services covered in AI-102—including language, vision, and document intelligence—remain in AI-103 and are covered in Learning Paths 3, 4, and 5. Your existing knowledge of those service APIs, lab patterns, and common student questions will be directly applicable.

The key additions in AI-103 are Learning Paths 1 and 2, which introduce Microsoft Foundry as the development platform and cover generative AI and agentic solution development using the Azure AI Foundry SDK and Azure AI Agent Service. Trainers new to these topics should prioritize hands-on time with Microsoft Foundry and the agent development labs before delivery. AI-102 instructors who are already comfortable with the language and vision modules can focus their preparation on these new Foundry and agent-focused learning paths.

Real-world use cases

The idea behind this section is to demonstrate to students how various features in the course are implemented in the real world. In other words, take the students from concepts covered in the training to actual scenarios involving course features.

Feedback

In this course, we have provided a framework for you to work with. Take time to prepare and think about the value that only an instructor can bring to training. We hope to partner with you to provide an exceptional student experience, and we welcome your feedback.